

in the January/February 2015 issue of the *Financial Analysts Journal*, and the “actuarial approach” described by Ken Steiner in the fall 2014 issue of the *Journal of Personal Finance* in an article named “A Better Systematic Withdrawal Strategy—the Actuarial Approach.” As well, members of the Bogleheads Forum collectively developed a variant of this approach described in various discussion threads and on their wiki, which they call “variable percentage withdrawal.”

These approaches recognize that the amount someone can spend in each year of retirement can be determined through a simple annuity calculation for a spending rate assuming a fixed portfolio return and remaining time horizon. Their calculations can be updated annually for the new portfolio balance, any changes to the expected return, an adjustment for remaining longevity, and other rules to limit spending adjustments.

There are also a number of other more sophisticated actuarial methods that incorporate Monte Carlo simulations to calibrate spending based on a specified probability of success or failure. David Blanchett from Morningstar has been involved with many of these efforts, and his joint work was distilled into a simple formula for retirement withdrawals that he published as “Simple Formulas to Implement Complex Withdrawal Strategies” in the September 2013 issue of the *Journal of Financial Planning*. His spreadsheet that accompanied the article allows users to input their preferred asset allocation, expected portfolio returns, level of portfolio fees, remaining life expectancy, and targeted probability of success, in order to obtain a customized withdrawal rate. The genesis of this work began in the age-based, three-dimensional distribution model he developed with Larry Frank and John Mitchell in a series of three articles published in 2011 and 2012 issues of the *Journal of Financial Planning*, with Larry Frank as the lead author. Though these methods are more sophisticated to allow for a probability of failure, the underlying PMT formula remains as the philosophical core of the spending recommendations.